POOJA KIRAN | SECURITY ENGINEERING

Evidence-Centered Visualization for ML Security Engineering

Aggregating Repository-Level Measurements Into Traceable Technical Dashboards

Source inventory: c651768630530d2f8f2f4ff5f96ebd53e2c44142

Test snapshot: c651768630530d2f8f2f4ff5f96ebd53e2c44142

CI run 38298823655 | initiated 2026-10-12 UTC | MIT

https://github.com/poojakira/mlsec-dashboards

37
tests passed in cited CI

8
default configured repositories, not confirmed aggregation

Token
REST API authentication in implemented path

JSON
recognized metrics from available sibling files

CONTRIBUTION

- An evidence-centered dashboard server that aggregates committed per-repo JSON into traceable HTML + a token-authed REST API, deliberately surfacing weak results (e.g. ROC-AUC 0.54, OOD F1 0.72) alongside strong ones rather than curating highlights.

IMPLEMENTED METHOD

- 1. Configure sibling repository locations.
- 2. Authenticate protected REST requests with the configured token.
- 3. Discover available JSON in supported evidence directories.
- 4. Extract recognized fields and return aggregate results.
- 5. Render static HTML examples separately; check each example against its cited inputs.

LIMITATIONS AND EVIDENCE BOUNDARY

- Static evidence snapshots, not live monitoring.
- Data can lag the current CI state of each repo.
- No alerting or real-time telemetry.
- Depends on sibling repos committing evidence.
- Local aggregation; not a hardened service.
- Not a live SOC monitor or hardened multi-user deployment; static snapshots can lag source CI.

SCOPED EVIDENCE

- Cited Python 3.12 CI: 37 passed. No coverage percentage is asserted.
- Eight repositories are configured by default; the number successfully aggregated depends on available sibling files.
- The server discovers JSON under evidence, results, metrics, reports, output and repository root, and extracts recognized shallow fields. Static authored dashboard examples are not all dynamically sourced.
- Prior sibling examples use dimensionless metrics: grouped F1 about 0.97, OOD F1 about 0.72, ROC-AUC about 0.54. These are example inputs, not new dashboard measurements.

REPRODUCTION

- git checkout c651768630530d2f8f2f4ff5f96ebd53e2c44142
- Install the documented development dependencies; run python -m pytest tests/ -q. Use the cited job configuration when reproducing coverage.
- Benchmark data, adapters and environment requirements are separate; consult poster/13_reproducibility.md and the claim ledger.

Dated tests and committed fixtures do not establish universal detection, production deployment or an SLO.